He ZiAng

System Engineer | Backend APIs & System Integration

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Professional summary

System Engineer developing and supporting manufacturing software since 2022, with a focus on backend services, system integration, and equipment-data workflows for engineering and quality teams.

Technical skills

Backend and APIs: Node.js, Express, Flask, Bun, ASP.NET, REST APIs.

Databases and integration: Oracle, PostgreSQL, SQL, MQTT, WebSockets, SPC integration.

Languages and data processing: Python, JavaScript, TypeScript, VB.NET, pandas, Parquet; earlier Swift and Java experience.

Delivery and interfaces: Git, Docker, React.

Work experience

Resonac HD Singapore 2022.03 - Present
System Engineer Singapore

- Developed Node.js/Express APIs backed by stored procedures for a measurement-data platform, supporting engineering configuration, data processing, and result review. Built ingestion, Excel reporting, and web-interface workflows.
- Built a visual inspection API and frontend with route permissions, transactional database writes, audit history, reporting, and inspection-image access. Prepared Docker-based UAT deployment and database integration.
- Developed equipment-log pipelines for multiple production lines, converting manufacturing logs into Parquet datasets, extracting process cycles, tracking metadata, and archiving data. Added database connection pooling, processing diagnostics, and visualization applications.
- Developed and supported event-driven monitoring for pump condition, motor torque, and process parameters. Implemented threshold alerts, PLC-connected cassette tracking, MQTT integration, WebSocket broadcasts, and exception-log ingestion.
- Extended an ASP.NET/VB.NET measurement-management system with instrument configuration, requests, acknowledgements, result uploads, and statistical process control (SPC) integration. Added instrument workflows and contributed to measurement reporting.
- Contributed to migrating manufacturing alert integrations from Workplace to Microsoft Teams. Updated application integrations, caller configuration, and message formatting through a shared Microsoft Graph service.
- Used LLM coding assistants, including Codex and Claude Code, for development, refactoring, debugging, and test preparation, with review and verification of generated changes.

Vanitee 2017.09 - 2018.02
iOS Developer Intern Singapore

- Worked on a Swift business application for iPhone and iPad and integrated React Native into the iOS application to support code sharing with Android.

Education

University of New South Wales 2018 - 2020
Computer Science (Artificial Intelligence) Bachelor of Science

Ngee Ann Polytechnic 2015 - 2018
Information Technology Diploma

Languages

Chinese [Native] / English [Professional]